

The Optimized Liu–Tang–Zhang Constant for Cactus Graphs

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Abstract

For a connected graph G , we optimize the doubly nonnegative functional underlying the square-energy theorem of Liu, Tang, and Zhang. If G is a cactus with b bridge blocks and cyclic blocks of lengths $\ell_1, \dots, \ell_r$, then its exact optimized constant is

$$\kappa(G) = b + \sum_{j=1}^r \kappa(C_{\ell_j}), \quad \kappa(C_\ell) = \begin{cases} \ell, & \ell \text{ even,} \\ \ell \sec^2(\pi/(2\ell)), & \ell \text{ odd.} \end{cases}$$

The proof combines a direct Fourier optimization of the Liu–Tang–Zhang functional on a cycle with exact additivity under a one-vertex sum. The latter requires both directions: folding and a Gram split give the upper bound, while orthogonal Gram gluing with the correct relative scaling gives the lower bound. Applied to the negative spectral part of the adjacency matrix, the result yields $s^-(G) \leq \kappa(G)$. For bicyclic cacti this settles the positive square-energy conjecture of Akbari–Kumar–Mohar–Pragada–Zhang except possibly when the unordered pair of cyclic block lengths is $\{5, 5\}$ or $\{3, 4k+1\}$, $k \geq 1$. These are unresolved families here, not asserted exceptions or counterexamples.

1 Introduction

All graphs are finite, simple, and undirected. If the adjacency eigenvalues of G are $\lambda_1, \dots, \lambda_n$, write

$$s^+(G) = \sum_{\lambda_i > 0} \lambda_i^2, \quad s^-(G) = \sum_{\lambda_i < 0} \lambda_i^2.$$

These quantities were introduced in connection with spectral bounds for the chromatic number by Elphick, Farber, Goldberg, and Wocjan [2]. Their conjecture $\min\{s^+, s^-\} \geq n - 1$ for connected graphs was recently proved by Liu, Tang, and Zhang [3]. The central device in their proof is an inequality for doubly nonnegative matrices, meaning real symmetric matrices that are both positive semidefinite and entrywise nonnegative.

We optimize that device graph by graph. For a connected graph with at least one edge, set

$$\kappa(G) = \sup_{\substack{M \succeq 0, \\ T(M) > 0}} \frac{4S_G(M)^2}{T(M)}, \quad S_G(M) = \sum_{uv \in E(G)} \sqrt{M_{uv}}, \quad T(M) = \mathbf{1}^T M \mathbf{1}. \quad (1)$$

Thus M ranges over the doubly nonnegative cone. The universal Liu–Tang–Zhang inequality says

$$\kappa(G) \leq 2|E(G)| - |V(G)| + 1.$$

Our purpose is not to replace their framework but to compute its sharp constant on cacti.

A *cactus* is a connected graph in which every block is a bridge or a cycle. Equivalently, any two cycles have at most one vertex in common. Our main theorem is the following.

Theorem 1.1. *Let G be a connected cactus with b bridge blocks and cyclic blocks $C_{\ell_1}, \dots, C_{\ell_r}$. Then*

$$\kappa(G) = b + \sum_{j=1}^r \kappa(C_{\ell_j}), \quad (2)$$

where

$$\kappa(C_\ell) = \begin{cases} \ell, & \ell \text{ is even,} \\ \ell \sec^2\left(\frac{\pi}{2\ell}\right), & \ell \text{ is odd.} \end{cases} \quad (3)$$

The word “sharp” in Theorem 1.1 refers to the optimized DNN certificate (1). It does not say that the spectral bound $s^-(G) \leq \kappa(G)$ is always an equality.

Akbari, Kumar, Mohar, Pragada, and Zhang conjectured that every connected n -vertex graph with at least $n + 1$ edges satisfies $s^+(G) \geq n$ [1, Conjecture 1.2]. The exact cactus constant proves this for a large class of bicyclic cacti. The remaining common-phase cases are supplied by a direct packing-two Sachs argument: $s^+ > s^-$ when every cycle has length 3 (mod 4) and at most two cycles can be pairwise vertex-disjoint. Combining the two methods gives an exact account of what is proved in this class; see Theorem 5.4.

2 The exact cyclic certificate

We evaluate the cyclic instance of the Liu–Tang–Zhang functional directly. Given a DNN matrix M , fold each nonedge entry onto the corresponding two diagonal entries:

$$\mathcal{F}_G(M) = M + \sum_{\substack{\{u,v\} \notin E(G) \\ u \neq v}} M_{uv}(e_u - e_v)(e_u - e_v)^T. \quad (4)$$

Each summand is positive semidefinite and entrywise nonnegative off the two cancelled positions. Consequently $\mathcal{F}_G(M)$ is DNN, vanishes on nonedges, has the same edge entries as M , and has the same value of T . This is the folding step in the LTZ argument.

Proposition 2.1. *For every $\ell \geq 3$,*

$$\kappa(C_\ell) = \begin{cases} \ell, & \ell \text{ even,} \\ \ell \sec^2(\pi/(2\ell)), & \ell \text{ odd.} \end{cases}$$

Proof. Fold M by (4), and then average it over all cyclic rotations. The resulting matrix $\overline{M}$ is circulant and DNN, while $T(\overline{M}) = T(M)$. If t is its common cycle-edge entry, concavity of the square root gives

$$S_{C_\ell}(\overline{M}) = \ell \sqrt{\frac{1}{\ell} \sum_{e \in E(C_\ell)} M_e} \geq S_{C_\ell}(M).$$

Write r for the common row sum of $\overline{M}$. Then

$$T(\overline{M}) = \ell r, \quad \frac{4S(\overline{M})^2}{T(\overline{M})} = \frac{4\ell t}{r}.$$

Suppose first that $\ell = 2m$. Let $x_d \geq 0$ be the circulant entry at cyclic distance d , so $x_1 = t$. The Fourier eigenvalue belonging to the alternating vector is

$$\lambda_m = x_0 + 2 \sum_{d=1}^{m-1} (-1)^d x_d + (-1)^m x_m \geq 0.$$

Comparing it with the row sum gives

$$r - \lambda_m = 4 \sum_{\substack{1 \leq d < m \\ d \text{ odd}}} x_d + 2 \mathbf{1}_{\{m \text{ odd}\}} x_m \geq 4t.$$

Hence $r \geq 4t$ and the objective is at most ℓ .

Now let $\ell = 2m+1$ and put $c = \cos(\pi/\ell)$. The Fourier eigenvalue at frequency m is nonnegative. Therefore

$$\begin{aligned} r - \lambda_m &= 2 \sum_{d=1}^m \left(1 - \cos \frac{2\pi m d}{\ell} \right) x_d \\ &\geq 2(1+c)x_1 = 2(1+c)t, \end{aligned}$$

where only the $d = 1$ term was retained and all other terms are nonnegative. Thus $r \geq 2(1+c)t$, and the objective is at most

$$\frac{2\ell}{1+c} = \ell \sec^2\left(\frac{\pi}{2\ell}\right).$$

For the reverse inequality let $A = A(C_\ell)$, choose $x > 0$, and put

$$M = x(2cI + A), \quad c = \begin{cases} 1, & \ell \text{ even}, \\ \cos(\pi/\ell), & \ell \text{ odd}. \end{cases} \quad (5)$$

The least eigenvalue of $A(C_\ell)$ is -2 for even ℓ and $-2\cos(\pi/\ell)$ for odd ℓ , so M is positive semidefinite. It is also entrywise nonnegative. Every edge entry is x , and

$$S = \ell\sqrt{x}, \quad T = 2x\ell(c+1).$$

Thus the objective is $2\ell/(c+1)$, which equals ℓ in the even case and $\ell \sec^2(\pi/(2\ell))$ in the odd case. This matches the certificate. $\square$

3 Exact one-vertex-sum additivity

Write $G = G_1 \vee_z G_2$ when G_1 and G_2 are connected edge-containing graphs whose intersection is exactly the vertex z , with no edges between $V(G_1) \setminus \{z\}$ and $V(G_2) \setminus \{z\}$.

Theorem 3.1 (One-vertex-sum additivity). *For $G = G_1 \vee_z G_2$,*

$$\kappa(G) = \kappa(G_1) + \kappa(G_2).$$

Proof. Put $\kappa_i = \kappa(G_i)$. For the upper bound, first fold a feasible M on G . Let g_v be Gram vectors for the resulting edge-supported matrix, and set

$$\mathcal{U} = \text{span}\{g_u : u \in V(G_1) \setminus \{z\}\}, \quad \mathcal{W} = \text{span}\{g_w : w \in V(G_2) \setminus \{z\}\}.$$

The nonedge zeroes imply $\mathcal{U} \perp \mathcal{W}$. Decompose $g_z = p_U + p_W + p_0$, where $p_U \in \mathcal{U}$, $p_W \in \mathcal{W}$, and $p_0 \perp \mathcal{U} + \mathcal{W}$. On G_1 use the original vectors away from z and the cut vector $p_U + p_0$ at z ; on G_2 use the original vectors away from z and p_W at z . Their Gram matrices M_1, M_2 are DNN, all edge inner products are preserved, and orthogonality gives

$$S_G(M) = S_{G_1}(M_1) + S_{G_2}(M_2), \quad T(M) = T(M_1) + T(M_2). \quad (6)$$

Writing $S_i = S_{G_i}(M_i)$ and $T_i = T(M_i)$, we obtain

$$2(S_1 + S_2) \leq \sqrt{\kappa_1 T_1} + \sqrt{\kappa_2 T_2} \leq \sqrt{(\kappa_1 + \kappa_2)(T_1 + T_2)}.$$

This proves $\kappa(G) \leq \kappa_1 + \kappa_2$.

For the lower bound, choose folded, edge-supported near-optimizers on the two graphs and represent them in orthogonal Euclidean spaces. Glue them by replacing the two vectors at z by their orthogonal direct sum; keep every other vector in its own summand. The resulting Gram matrix is DNN on G , has no cross-side entries, and again satisfies (6). Scale the two input matrices so that

$$\frac{T_1}{\kappa_1} = \frac{T_2}{\kappa_2}. \quad (7)$$

For exact optimizers this makes both Cauchy–Schwarz inequalities above equalities and gives objective value $\kappa_1 + \kappa_2$. Applying the same scaling to arbitrarily close near-optimizers and taking a limit proves the claim without assuming that either supremum is attained. $\square$

Proposition 3.2. $\kappa(K_2) = 1$.

Proof. For a DNN matrix $\begin{pmatrix} a & c \\ c & d \end{pmatrix}$, positive semidefiniteness gives $a+d \geq 2c$, and hence $T = a+d+2c \geq 4c$. Therefore the objective is at most one. Equality holds for any positive multiple of the all-ones matrix J_2 . $\square$

Every cactus is obtained by iterated one-vertex sums of its blocks. Applying Theorem 3.1, Proposition 3.2, and Proposition 2.1 proves Theorem 1.1.

4 The spectral consequence

Let $A = A(G)$ and write its positive and negative spectral parts as

$$A = P - B, \quad P = A_+ \succeq 0, \quad B = A_- \succeq 0, \quad PB = 0.$$

Thus $s^-(G) = \text{tr } B^2$. The next consequence is valid for every connected graph, with its own optimized constant.

Theorem 4.1. *If G is connected and has at least one edge, then*

$$s^-(G) \leq \kappa(G).$$

Proof. The Schur product theorem makes $M = B \circ B$ doubly nonnegative. Since A has two unit entries for every edge and $PB = 0$,

$$\text{tr}(AB) = \text{tr}((P - B)B) = -\text{tr } B^2 = -s^-(G),$$

and therefore

$$\frac{s^-(G)}{2} = - \sum_{uv \in E(G)} B_{uv} \leq \sum_{uv \in E(G)} |B_{uv}|.$$

Moreover, $T(B \circ B) = \sum_{u,v} B_{uv}^2 = \text{tr } B^2 = s^-(G)$. The defining DNN inequality for $\kappa(G)$ gives the chain

$$s^-(G)^2 \leq 4 \left(\sum_{uv \in E(G)} |B_{uv}| \right)^2 \leq \kappa(G) \mathbf{1}^T (B \circ B) \mathbf{1} = \kappa(G) s^-(G). \quad (8)$$

Since G has an edge, $s^-(G) > 0$, and division proves the result. $\square$

Remark 4.2. Even when the cactus formula for $\kappa(G)$ is attained by a DNN matrix, equality need not hold in the first inequality of (8), nor must $B \circ B$ optimize the DNN functional. This is why DNN sharpness must not be interpreted as spectral equality.

5 Bicyclic cacti

Define the odd-cycle excess

$$\epsilon_\ell = \begin{cases} 0, & \ell \text{ even,} \\ \ell \tan^2(\pi/(2\ell)), & \ell \text{ odd.} \end{cases} \quad (9)$$

Then $\kappa(C_\ell) = \ell + \epsilon_\ell$.

Lemma 5.1 (Exact trigonometric comparison). *The sequence ϵ_ℓ is strictly decreasing through odd integers $\ell \geq 3$, and*

$$\epsilon_3 = 1, \quad \epsilon_5 = 5 - 2\sqrt{5}, \quad \epsilon_5 + \epsilon_7 < 1.$$

Proof. Put $t = \pi/(2x)$. Apart from the positive constant $\pi/2$, $x \tan^2(\pi/(2x))$ is $\tan^2 t/t$. Its derivative with respect to t is positive because

$$2t \tan t \sec^2 t - \tan^2 t > 0 \iff 2t > \sin t \cos t,$$

which holds on the relevant interval $0 < t \leq \pi/6$. Since t decreases with x , the required monotonicity follows. The first identity is immediate, and $\tan^2(\pi/10) = 1 - 2\sqrt{5}/5$ gives the second.

For a fully strict comparison in the last assertion, $\pi < 22/7$ and $\cos x > 1 - x^2/2$ give

$$\cos(\pi/7) > 1 - \frac{1}{2} \left(\frac{22}{49} \right)^2 > \frac{7}{8}.$$

Hence

$$\epsilon_7 = 7 \frac{1 - \cos(\pi/7)}{1 + \cos(\pi/7)} < \frac{7}{15} < 2\sqrt{5} - 4 = 1 - \epsilon_5.$$

For the middle strict inequality, $7/15 < 2\sqrt{5} - 4$ is equivalent to $67/30 < \sqrt{5}$, whose square follows from $4489 < 4500$. $\square$

Let G now be a connected bicyclic cactus, and let p, q be the lengths of its two cyclic blocks. Since $|E(G)| = |V(G)| + 1$, if $n = |V(G)|$ and b is the number of bridges, then $b + p + q = n + 1$. Theorem 1.1 and Theorem 4.1 therefore give

$$s^-(G) \leq n + 1 + \epsilon_p + \epsilon_q. \quad (10)$$

Since $s^+(G) + s^-(G) = \text{tr } A^2 = 2|E(G)| = 2n + 2$, we obtain the following direct criterion.

Corollary 5.2. *If $\epsilon_p + \epsilon_q \leq 1$, then $s^+(G) \geq n$. In particular, this holds whenever at least one of p, q is even, and also whenever p, q are odd with $\min\{p, q\} \geq 5$ and $\{p, q\} \neq \{5, 5\}$.*

Proof. Equation (10) gives $s^+(G) \geq n + 1 - \epsilon_p - \epsilon_q \geq n$. Even lengths have zero excess, while every odd excess is at most $\epsilon_3 = 1$, proving all cases with an even length. For two odd lengths at least five, the only pair not bounded by $\epsilon_5 + \epsilon_7 < 1$ and monotonicity is $\{5, 5\}$. $\square$

The DNN estimate does not settle pairs containing a triangle and another odd cycle, because $\epsilon_3 = 1$ and the second excess is positive. When the second cycle also has length 3 (mod 4), the independent phase method does more than is needed.

Proposition 5.3 (Packing-two consequence). *If $p \equiv q \equiv 3 \pmod{4}$, then*

$$s^+(G) > s^-(G), \quad \text{and consequently} \quad s^+(G) > |E(G)| = n + 1.$$

Proof. Let

$$Z_H(t) = \sum_j m_j(H) t^{|V(H)|-2j} > 0 \quad (t > 0)$$

be the signless matching polynomial. Grouping the Sachs expansion by its collection $\mathcal{C}$ of vertex-disjoint cycle components gives

$$i^{-|V(G)|} \phi_G(it) = \sum_{\mathcal{C}} \prod_{C \in \mathcal{C}} (-2i^{-|C|}) Z_{G-V(C)}(t).$$

A bicyclic cactus has only two cyclic blocks, so $|\mathcal{C}| \leq 2$. If both lengths are 3 (mod 4), every singleton cycle has phase $-2i$, while the empty and two-cycle terms are real. Hence the normalized characteristic polynomial lies strictly in the open lower half-plane for every $t > 0$. Its continuous argument

$$\Theta_G(t) = \sum_j \arctan(\lambda_j/t)$$

is therefore negative. The signed Coulson identity

$$s^+(G) - s^-(G) = -\frac{4}{\pi} \int_0^\infty t \Theta_G(t) dt$$

yields $s^+ > s^-$. The energy identity then turns this into $s^+ > |E(G)|$. $\square$

We can now state the broad classification supplied by the two methods.

Theorem 5.4 (Bicyclic cactus classification). *Let G be a connected bicyclic cactus on n vertices, with unordered pair of cyclic block lengths $\{p, q\}$. Then $s^+(G) \geq n$ in all cases except possibly*

$$\{p, q\} = \{5, 5\} \quad \text{or} \quad \{p, q\} = \{3, 4k + 1\} \quad (k \geq 1). \quad (11)$$

More explicitly:

1. *if at least one length is even, the optimized DNN bound proves the conjecture;*
2. *if both lengths are odd and at least five, the optimized DNN bound proves it unless both are five;*
3. *if both lengths are 3 (mod 4), the packing-two Sachs argument proves the stronger inequality $s^+(G) > |E(G)| = n + 1$;*

4. after these cases, exactly the pairs in (11) remain.

In particular, the conjecture of Akbari–Kumar–Mohar–Pragada–Zhang holds for every bicyclic cactus outside the two displayed families.

Proof. Corollary 5.2 handles all even cases and all odd pairs with both lengths at least five except $\{5, 5\}$. It remains to inspect pairs containing a triangle. If the other odd length is $3 \pmod{4}$, including another triangle, Proposition 5.3 applies. The only untreated odd partner is therefore $4k + 1$ with $k \geq 1$. This gives precisely (11). $\square$

Remark 5.5 (Status of the residual families). The word “possibly” is essential. Neither $\{5, 5\}$ nor $\{3, 4k + 1\}$ is claimed to be an exception to the AKMPZ conjecture, and no counterexample is asserted. They are exactly the families not resolved by the optimized cactus DNN certificate together with the cited packing-two theorem. Arbitrary bridge paths, shared cut vertices, and attached trees are already included in this statement.

References

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